

LostFound App - User Manual

A campus lost-and-found reporting system with real-time TCP communication.

Prerequisites

For the Backend Server (Python)

- Python 3.8 or higher
- pip (Python package manager)

For the Android App

- Android Studio (latest version recommended)
 - JDK 11 or higher
 - Android SDK (API level 28 minimum, targeting API 35)
 - An Android device or emulator running Android 9.0 (Pie) or higher
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Setup Instructions

Step 1: Install Python Dependencies

Open a terminal in the project root directory (`tcp/`) and run:

```
pip install supabase
```

The server uses the following Python packages:

- `supabase` - For database and storage operations
- `socket` , `threading` , `base64` , `uuid` , `datetime` - Standard library (no installation needed)

Step 2: Start the Backend Server

From the project root directory, run:

```
python server.py
```

The server will start listening on:

- **Host:** `0.0.0.0` (accepts connections from any interface)
- **Port:** `12345`

You should see output indicating the server is running.

Step 3: Build and Run the Android App

Option A: Using Android Studio (Recommended)

1. Open Android Studio
2. Select **File > Open** and navigate to `tcp/LostFound/`
3. Wait for Gradle sync to complete
4. Connect your Android device via USB or start an Android emulator
5. Click the **Run** button (green play icon) or press `Shift+F10`

Install on a connected device:

```
adb install app/build/outputs/apk/debug/app-debug.apk
```

Network Configuration

Important: Server IP Address

The Android app needs to connect to the server. By default, the server runs on your local machine.

For Emulator Testing:

- Use `10.0.2.2` as the server IP (Android emulator's alias for host machine)

For Physical Device Testing:

- Use your computer's local IP address (e.g., `192.168.1.x`)
- Ensure both devices are on the same network
- Check your IP with:
 - Windows: `ipconfig`
 - macOS/Linux: `ifconfig` or `ip addr`

Firewall Configuration

Ensure port `12345` is open on your firewall for incoming TCP connections.

App Features

Feature	Description
User Authentication	Register and login to your account
Report Creation	Report lost or found items with photos
View Reports	Browse all lost and found reports
My Reports	Manage your own reports
Real-time Chat	Message other users about items
Notifications	Receive updates on your reports
Location Sharing	Share location for meetups
Profile Management	Update your profile information
Claims System	Claim items and manage claims
Photo Viewing	View item photos in detail

Required Permissions

The app requires the following permissions:

- **Internet** - For server communication
 - **Camera** - To take photos of items
 - **Storage** - To select photos from gallery
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Quick Start Summary

1. Install Python dependency: `pip install supabase`
2. Start server: `python server.py`
3. Open `LostFound/` in Android Studio
4. Configure server IP in the app (if needed)
5. Run the app on device/emulator
6. Register a new account and start using the app